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Mason et al.

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(54) **COMPACT GASIFIER-GENSET
ARCHITECTURE**

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(52) **U.S. Cl.**
USPC **290/1 A**

(58) **Field of Classification Search**
USPC 290/1 R, 1 A, 2; 60/648
See application file for complete search history.

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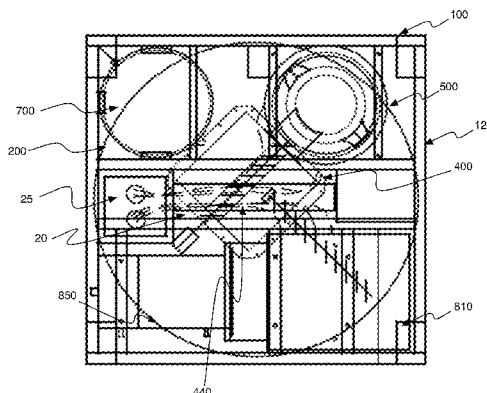
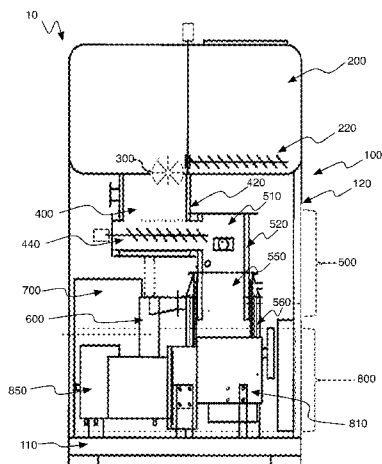
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(57) **ABSTRACT**

A compact biomass gasification-based power generation system that converts carbonaceous material into electrical power, including an enclosure that encases: a gasifier including a pyrolysis module coaxially arranged above a reactor module, a generator including an engine and an alternator, and a hopper. The generator system additionally includes a first heat exchanger fluidly connected to an outlet of the reactor module and thermally connected to the drying module, a second heat exchanger fluidly connected to an outlet of the engine and thermally connected to the pyrolysis module, and a third heat exchanger fluidly connected between the outlet of the reactor module and the first heat exchanger, the third heat exchanger thermally connected to an air inlet of the reactor module. The system can additionally include a central wiring conduit electrically connected to the pyrolysis module, reactor module, and engine, and a control panel connected to the conduit that enables single-side operation.

25 Claims, 7 Drawing Sheets



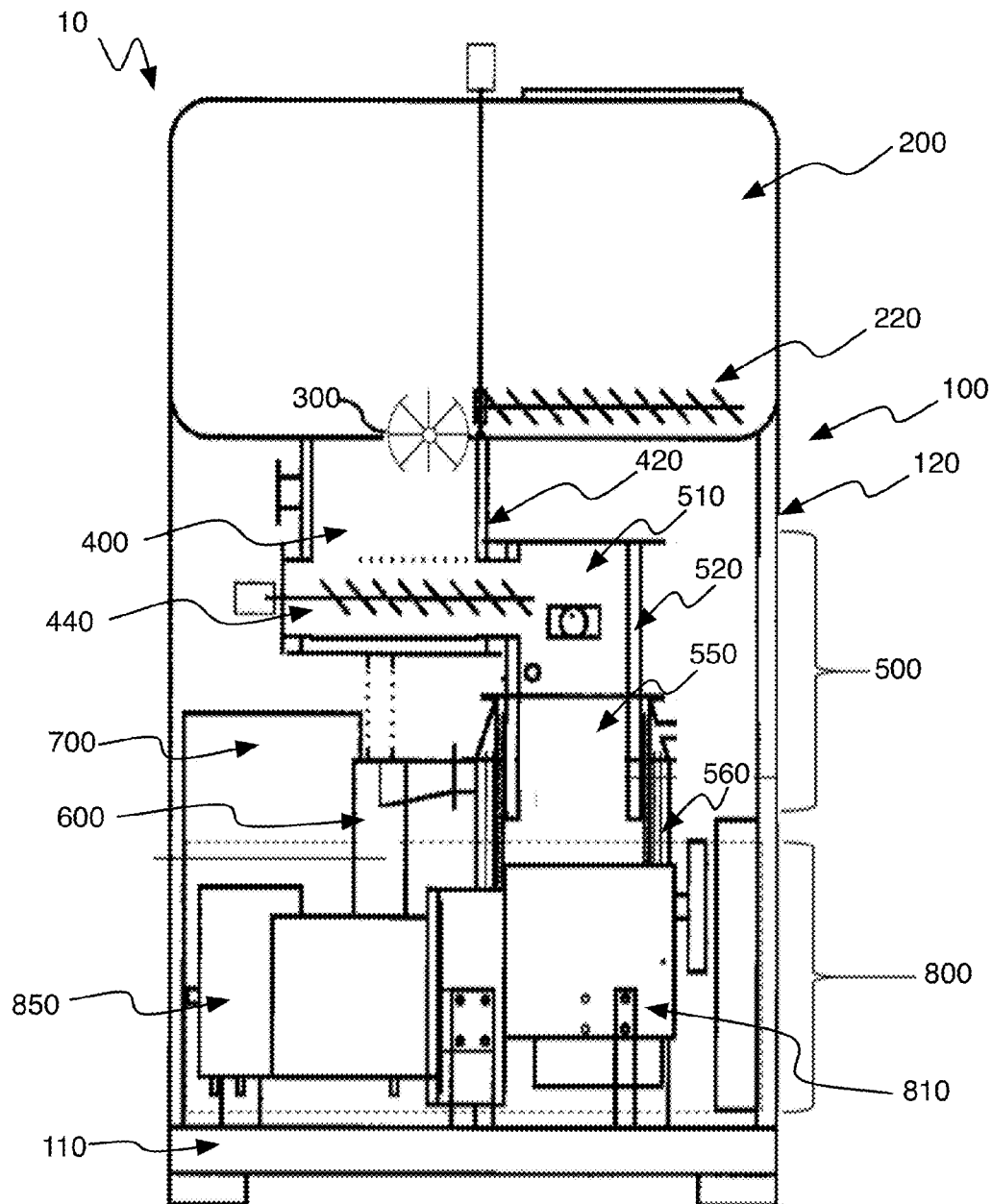


FIGURE 1A

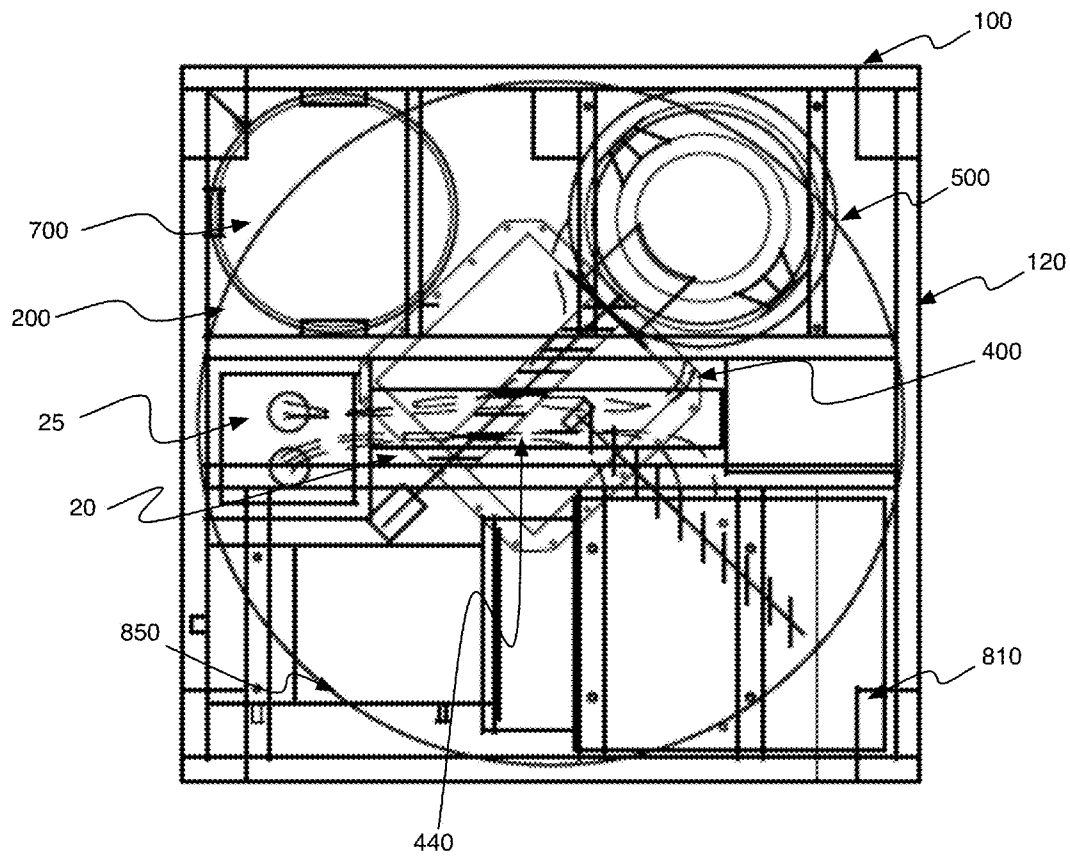
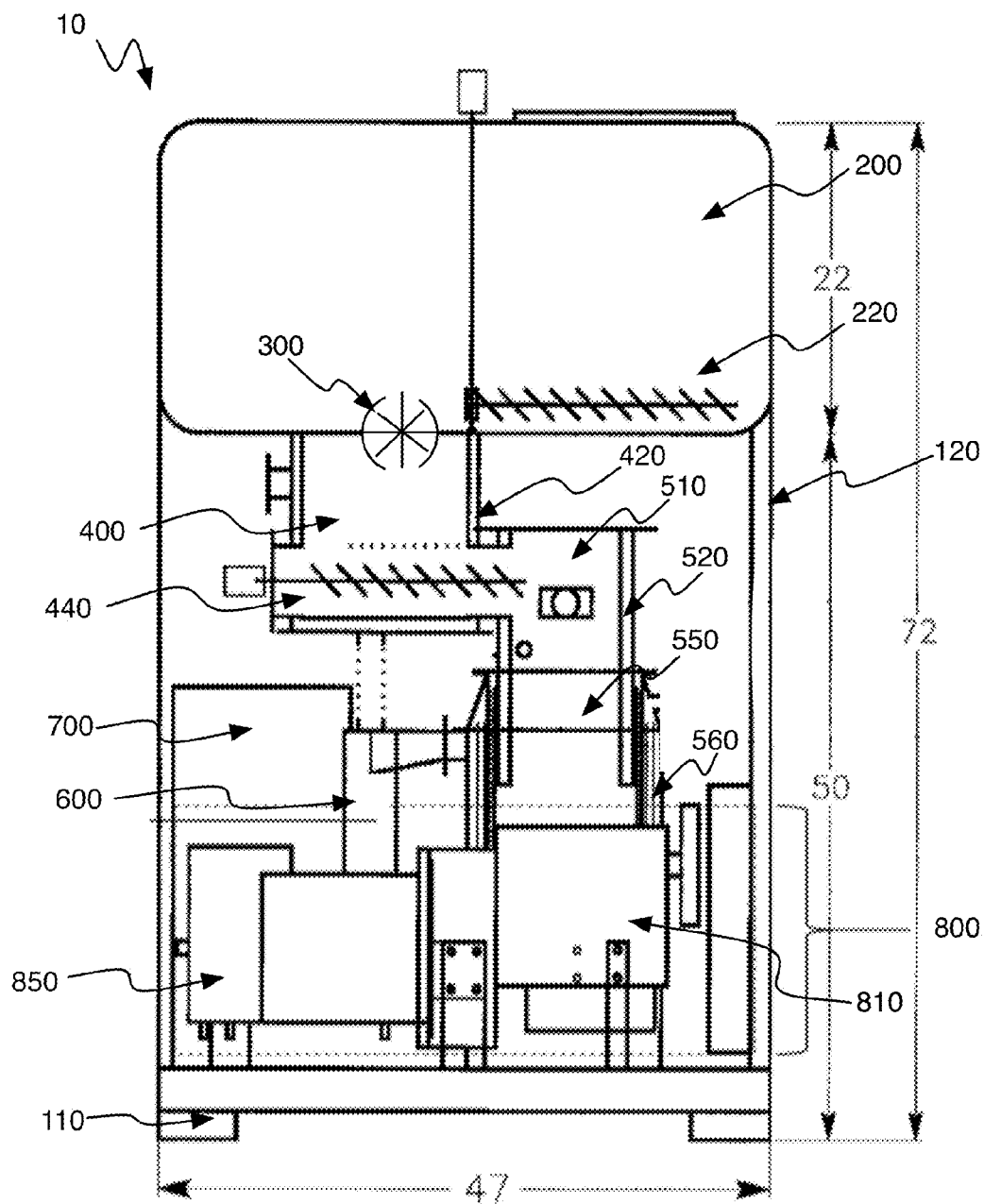


FIGURE 1B



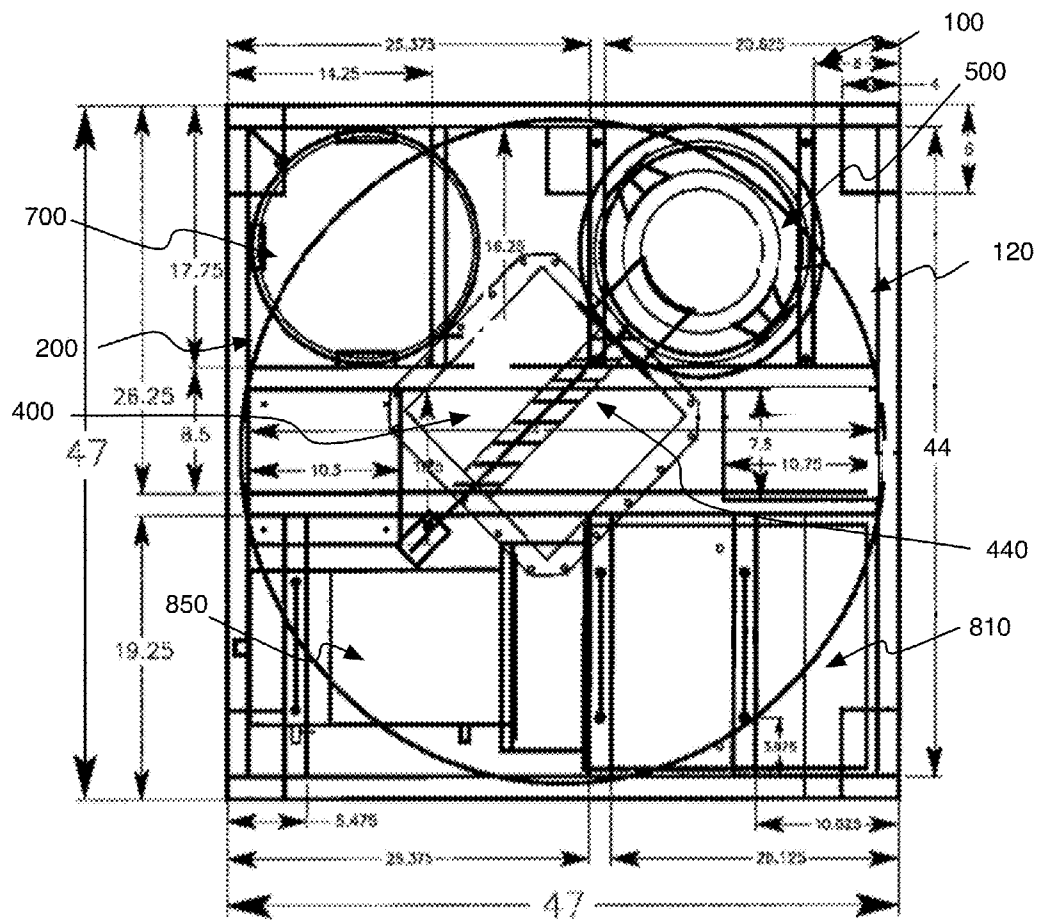


FIGURE 2B

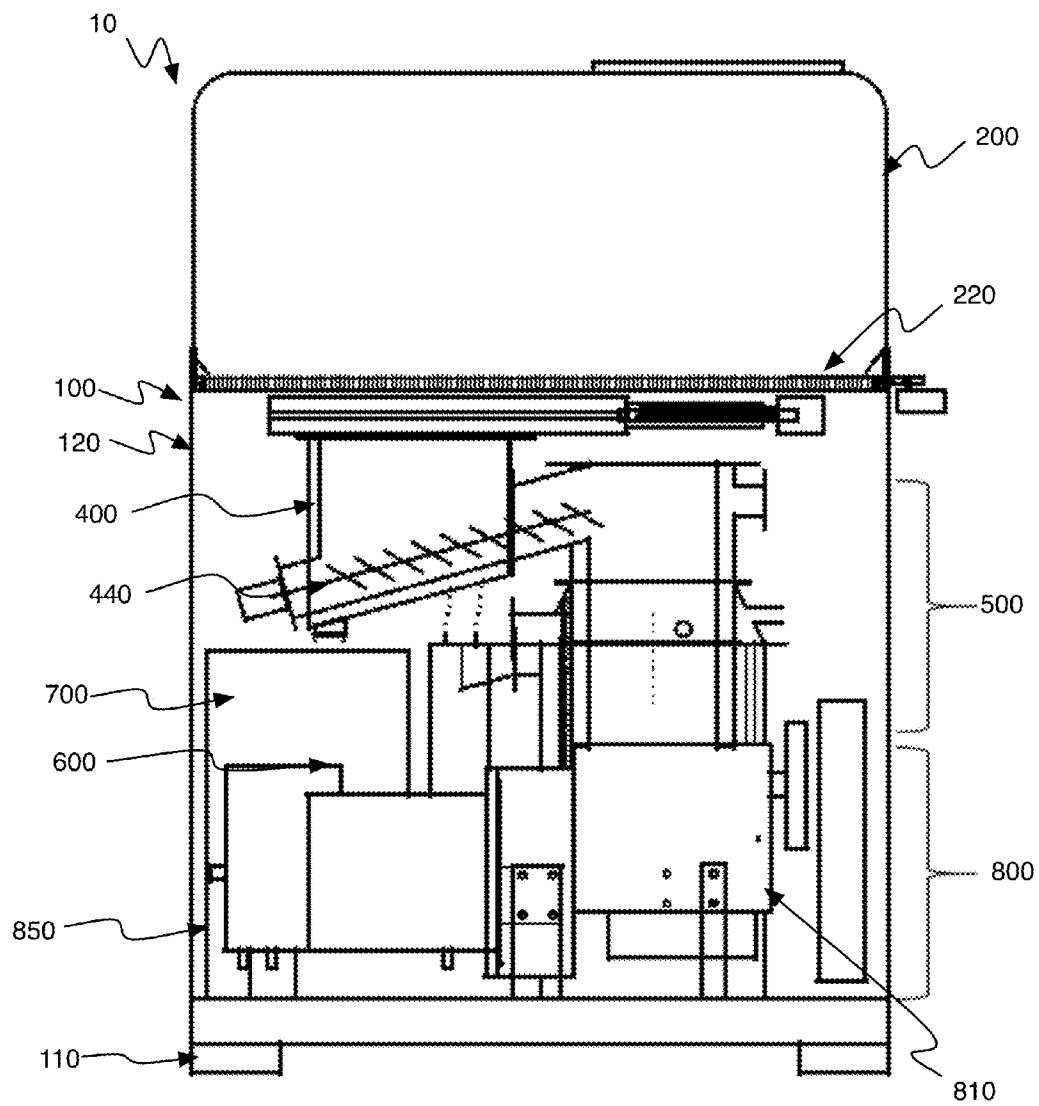


FIGURE 3A

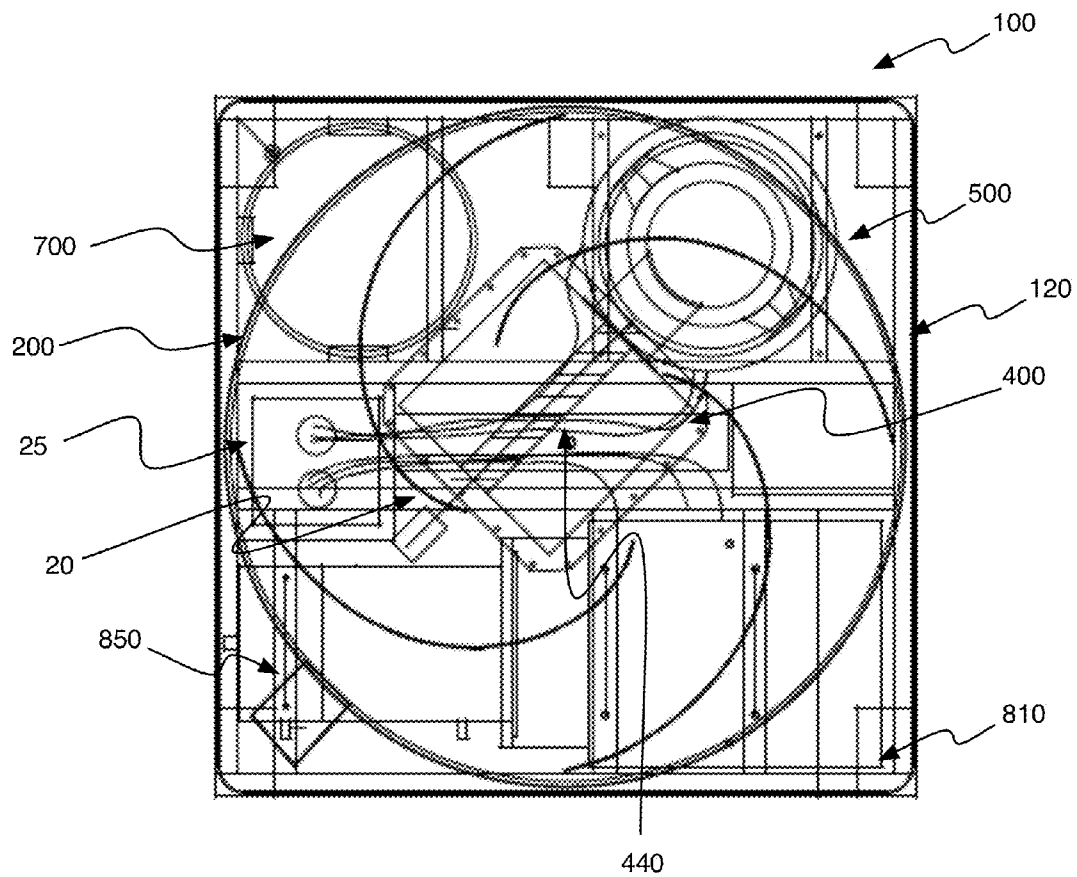


FIGURE 3B

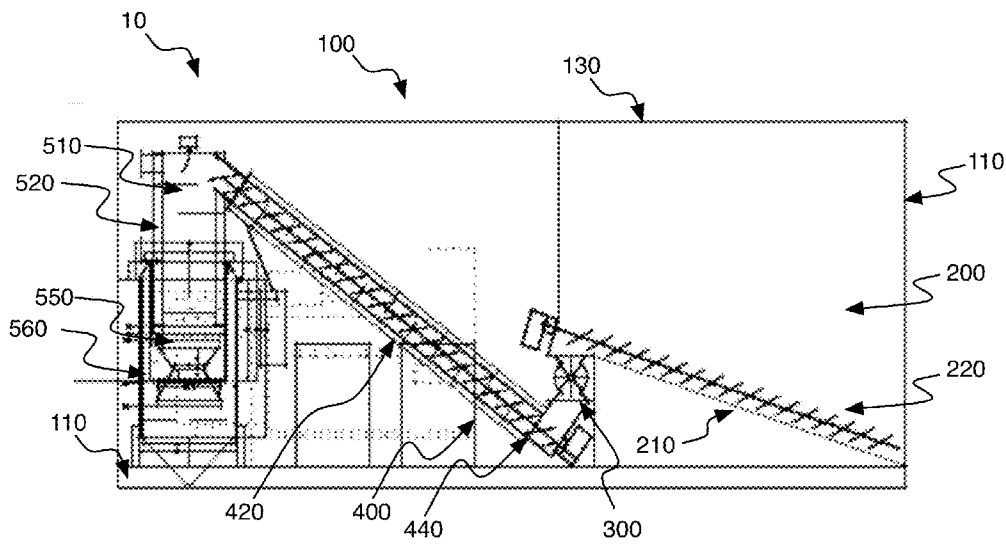


FIGURE 4A

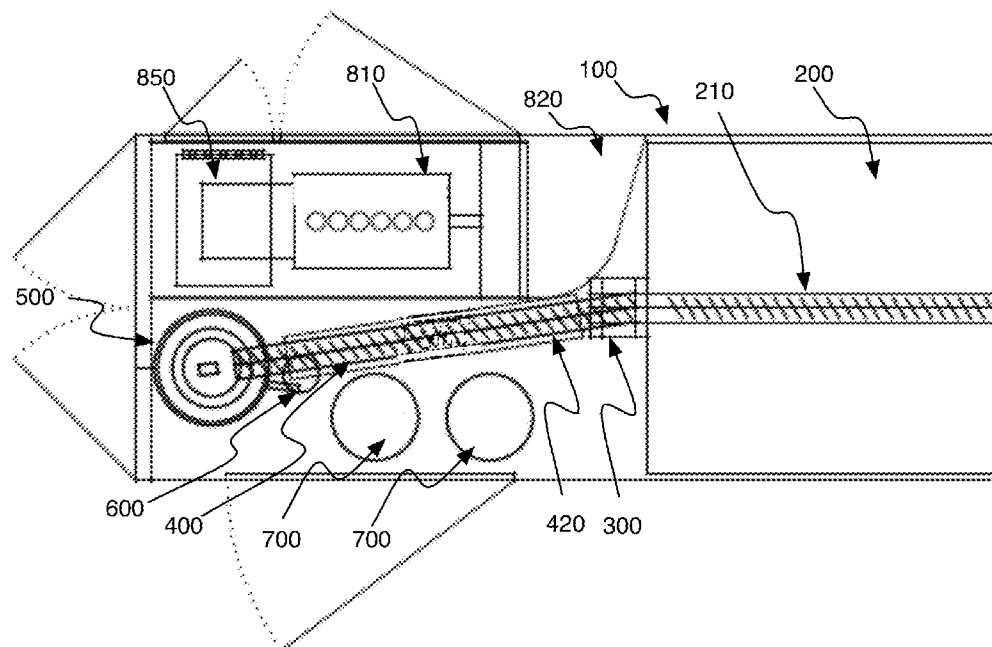


FIGURE 4B

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COMPACT GASIFIER-GENSET ARCHITECTURE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims benefit to U.S. Provisional No. 61/617,658 filed 29-MAR-2012, which is incorporated in its entirety by this reference.

This application is related to U.S. patent application Ser. No. 12/846,807, filed 29 Jul. 2010 and titled "SYSTEM AND METHOD FOR DOWNDRAFT GASIFICATION," which is incorporated in its entirety by this reference.

TECHNICAL FIELD

This invention relates generally to the power generation field, and more specifically to an improved biomass gasification based power generator in the power generation field.

BACKGROUND

Biomass gasification based generators are an attractive method for power generation, given they are fueled with carbonaceous byproducts that are inexpensive or free to source. However, conventional biomass gasifiers are often large, sprawling assemblies, preventing consumer adoption due to their significant footprint, complexity of operation, and difficult transport logistics. What is desired is a complete and compact biomass power generation system in a single box enclosure, in the manner typical of a liquid-fueled genset, such as a diesel or LPG generator. However, it is difficult to achieve this configuration with a gasifier genset given the significant number of components involved, the large hopper required for low energy density biomass, and the complexity of solid fuel feeding systems. Until these problems are solved, very few biomass gasification based generators will be deployed, and their many potential benefits will not be realized. Therefore, there is a need in the biomass gasification generator field to create an improved, fully integrated, and compact system that meets user expectations for contemporary power generation equipment.

BRIEF DESCRIPTION OF THE FIGURES

FIGS. 1A and 1B are a side and top view of a variation of the power generation system, respectively.

FIGS. 2A and 2B are a side and top view of a specific example of the power generation system, respectively.

FIGS. 3A and 3B are a side and top view of an example of the power generator system with a swirl hopper material transporter.

FIGS. 4A and 4B are a side and top view of a second variation of the power generation system, respectively.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

The following description of the preferred embodiments of the invention is not intended to limit the invention to these preferred embodiments, but rather to enable any person skilled in the art to make and use this invention.

1. Gasifier-Engine Generator System

As shown in FIG. 1, the improved power generation system **10** includes a skid **110**, a hopper **200**, a drying module **400**, a

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gasifier **500** including a pyrolysis module **510** and a reactor module **550**, a particulate separator **600**, a filtration module **700**, and an engine module **800** that cooperate to produce gaseous fuel (syngas or producer gas) from carbonaceous material, and to extract particulates from the gaseous fuel. The gasifier **500** is preferably a downdraft gasifier, but can be any other suitable gasifier. This power generation system **10** is preferably arranged to leverage gravity in transferring fuel precursors to different processing modules, and to leverage various waste heat streams to improve gasifier performance. Furthermore, the arrangement allows the power generation system **10** to have a substantially small footprint (preferably rectangular, more preferably square), such that it can fit on a shipping pallet or within a shipping container. This arrangement also preferably shortens the piping connections between different modules, and allows for control valves to be localized in one area, more preferably on one side of the power generation system **10**, for ease of user operation. In one variation of the power generation system **10**, the electronics (e.g., engine module), fillers, and engine-flare valves (e.g., valves that switch fuel flow between the flare to the engine) are arranged along a common edge of the base no to simplify user operation. This arrangement can also allow the power generation system **10** to be shipped in a substantially ready-to-use arrangement, with little to no additional assembly required by the consumer.

The skid **110** of the power generation system **10** functions to support the components of the power generation system **10**. The skid **110** is preferably a portion of an enclosure **100**, which functions to mechanically protect the components of the power generation system **10**. However, the power generation system **10** can alternatively only include the skid **110**, and not include walls or a top. The enclosure **100** can additionally function to trap and direct the radiator blast upwards toward and potentially through the hopper **200**. The enclosure **100** preferably includes walls extending perpendicularly from the skid **110**. The enclosure **100** can additionally include a top **130** arranged parallel the skid **110** across the walls **120**. The skid **110** is preferably rectangular or square, with a first corner adjacent a second corner, a third corner adjacent the second corner, and a fourth corner adjacent the first and third corners, but can have any suitable geometric shape. The skid **110** preferably defines the power generation system footprint, but the system footprint can be otherwise defined. In one variation, the skid no is a pallet having standardized dimensions, more preferably standardized shipping pallet dimensions. The enclosure **100** preferably includes four walls, each extending from a respective skid edge, but can alternatively include any suitable number of walls, preferably one wall for each edge. The skid **110** and top can be substantially solid, be porous, be solid with a porous segment, be made of a plurality of tubes or planks, or have any other suitable construction or configuration. The walls **120** are preferably substantially solid, but can alternatively be porous, made of multiple tubes or planks, or have any other suitable construction. The top and walls are preferably removable from the base, but can alternatively be substantially permanently coupled to the base (e.g., welded to the base). The top and walls can alternatively articulate relative to the base (e.g., be coupled by hinges to the base). The enclosure **100** can be a standardized shipping container (e.g., a standardized 20-ft intermodal container, 40-ft high-cube intermodal container, or any other suitable shipping container, etc.), a container built around a standardized base (e.g., a standardized 48 in×48 in pallet or any other suitable standardized pallet), or a container having any other suitable dimensions. The enclosure **100** is preferably metal (e.g., steel, a corrosion-resistant metal, or any other suitable

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metal), but can alternatively be made of wood, plastic, or any other suitable material. The enclosure 100 is preferably substantially rigid and opaque, but can alternatively be clear. The enclosure 100 is preferably made from an extruded metal sheet, bent and welded into a cube, but can be otherwise manufactured. The enclosure 100 is preferably encases the drying module 400, the gasifier 500 including the pyrolysis module 510 and the reactor module 550, and the particulate separator 600. The enclosure 100 can additionally enclose the airlock 300 if an airlock is used. The enclosure 100 can additionally enclose the engine module 800, the filtration module 700, and the control system. The enclosure 100 is preferably allows free air flow between the ambient environment and the enclosure interior, but can alternatively hold an air seal. The enclosure 100 can additionally enclose the hopper 200, but the hopper 200 can alternatively be mounted to enclosure 100 prior to generator use, wherein the hopper 200 is preferably mounted to the end of the walls 120 distal the skid 110 (e.g., by nuts and bolts coupling a flange of the hopper to the enclosure 100) but can alternatively be mounted to any suitable portion of the enclosure 100. The enclosure 100 can enclose a collapsed hopper, an extended hopper, a rigid hopper, cooperatively define the hopper with one or more auxiliary walls, or enclose any other suitable hopper.

The carbonaceous material fed into the power generation system 10 can be any suitable material that contains carbon, for example, biomass (e.g., wood, plants, algae), biodegradable waste (e.g., any waste generated by a plant or animal), or coal. The carbonaceous material is preferably fed into the hopper 200 by the user, but can alternatively be fed into the hopper 200 by a mechanized system or in any other suitable manner. In the power generator system 10, the carbonaceous material is converted to engine-grade gaseous fuel. The carbonaceous material is first dried by the drying module 400 and pyrolyzed into tar gas and charcoal in the pyrolysis module 510. The tar gas is then combusted and reduced within the reactor module 550 to create gaseous fuel, which is then passed through the air preheating- and gas cooling-heat exchanger 560, exits into the particulate separator 600 for particulate separation, is fed into the walls of the drying module 400 to dry the incoming carbonaceous material, is filtered in the filtration module 700, and is then fed into the carburetor of the engine module 800.

The power generator system 10 recaptures waste heat from both the gasifier and engine outputs, thereby simultaneously improving gasifier performance and reducing component count by cooling the gasifier output without a dedicated cooling module. In particular, the hot, gaseous fuel produced by the reactor module 550 preheats the air used for combustion within the reactor module 550 and also dries and preheats the carbonaceous fuel in the drying module 400, the hot exhaust gas from the engine 810 is used to heat the pyrolysis module 510; and the radiator blast from the engine module 800 is used for the first stage of drying the carbonaceous material in the hopper 200.

The hopper 200 of the power generation system 10 functions to receive carbonaceous material and can function to facilitate moisture removal from the carbonaceous material. The hopper 200 is preferably open to the ambient environment, but can alternatively be sealed, or have the ability to hold an air seal. In a first variation of the hopper 200, the hopper 200 preferably includes a carbonaceous material inlet that allows carbonaceous material to enter the hopper 200 and a hopper boot that allows carbonaceous material to egress into the drying module 400. The carbonaceous material inlet is preferably the open top of the hopper 200, but can alternatively be a channel, pipe, flume any other suitable lead-in in

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the side or base of the hopper 200. The hopper boot is preferably a controlled opening on the bottom of the hopper 200, and is preferably located above the drying module inlet, more preferably substantially near the center of the hopper base 210, but alternatively in any suitable position. The hopper 200 is preferably a generally cylindrical structure, and preferably has a diameter substantially similar to the width of the skid (shipping pallet) or enclosure footprint, but can alternatively be conical or any other suitable shape. An end of the hopper 200 is preferably fastened to the inlet of the drying module 400 (e.g., with nuts and bolts or any other suitable fastener), such that the hopper 200 is coupled to the drying module 400 during shipping or transportation, but the hopper 200 can alternatively be removably coupled to the drying module 400, be a separate component that is shipped in the same enclosure, or can be a component that the user assembles and installs. The hopper 200 is preferably made of a flexible material, preferably supported by ribbing (e.g., rings or a spiral along the height of the hopper 200), such that it can collapse and be transported with the body of the power generation system 10. Alternatively, the hopper 200 can be formed from multiple pieces of hinged material that can collapse into a substantially flat configuration for shipping, but can be extended to form a substantially cylindrical hopper 200. Alternatively, the hopper can be made of solid material, without collapsible features, and be shipped in its fully distended and operational form. The hopper material is preferably porous to allow better carbonaceous material drying, wherein the pores are preferably smaller than the carbonaceous material particulate size. In one variation, the hopper 200 includes a perforated base that facilitates carbonaceous material pre-drying with radiator heat from the engine 810 or particulate separator 600. The hopper material is preferably polymeric and corrosion-resistant, but can alternatively be treated, coated, or be bare metal. The hopper 200 is preferably formed from an extruded piece of material, but can alternately be formed from one or more pieces of woven, injection molded, cast, sintered, or any other suitable material manufactured in any suitable manner. The hopper 200 is preferably substantially transparent such that the user can see the fuel level, but can be semi-transparent or substantially opaque. However, any other suitable hopper in any suitable configuration that receives carbonaceous material can be utilized. In a second variation of the hopper, the hopper 200 is preferably substantially defined by the walls of the enclosure 100. More preferably, the hopper 200 occupies and is at least partially defined by an end of the enclosure 100 (e.g., the hopper 200 extends along approximately half of the enclosure longitudinal axis). The hopper 200 preferably includes a flat bottom, but can alternatively include an angled bottom, wherein the lower portion is preferably adjacent the enclosure end, or have any other suitable geometry. The hopper base 210 is preferably louvered or perforated to promote carbonaceous material drying, but can alternatively be solid. In one alternative of the hopper 200, the engine radiator 820 or particulate separator 600 extends under the raised portion of the angled hopper base 210. The hopper 200 can additionally include an external material transporter that moves carbonaceous material from the enclosure exterior to the hopper interior. The external material transporter preferably moves carbonaceous material from an area proximal the skid 110 to the top of the hopper 200, but can alternatively move carbonaceous material to and from any suitable position.

The hopper 200 can additionally include a carbonaceous material transporter 220 that moves the carbonaceous material from the body of the hopper 200 to the drying module 400, preferably through the airlock 300 but alternatively through

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an opening in the bottom of the hopper **200**. The carbonaceous material transporter **220** preferably removes the carbonaceous material from the bottom of the hopper (e.g., the portion of the hopper proximal the drying module **400** but alternatively the portion of the hopper **200** proximal the skid **110**), but can alternatively remove any suitable portion of the carbonaceous material within the hopper **200**. The carbonaceous material transporter **220** is preferably substantially parallel to the hopper base **210**, but can alternatively be at an angle to the hopper base **210**. The carbonaceous material transporter **220** is preferably a full bottom feeder, such as a sweep auger, scroll feeder, vibratory feeder, rotary table feeder, chain feeder, or any suitable material transporter. Utilizing a full bottom feeder can confer several benefits: first, it allows for a substantially cylindrical hopper to be used instead of a tapered hopper, which suffers from low hopper volume, failures with large and/or inconsistent fuels, and the bridging and arching modes of taper feeders. Second, a full bottom feeder allows for a low and wide hopper to be used, which allows for easier transportation and use (e.g., the hopper **200** is easier to fill). The full bottom feeder can include a sweep auger, scroll feeder, a rotary feed table, rotary plow feeder, vibratory feeder, flat bottom feeder, any suitable transporter that feeds solid fuel from a flat bottom container to single outlet chute. The rotation speed of the carbonaceous material transporter **220** can be varied depending on the fuel requirement of the gasifier **500** the existing carbonaceous material content within the gasifier **500**, and/or any other suitable factor. The carbonaceous material transporter **220** can be hand driven, but can alternatively be coupled to a motor that automatically rotates the auger to regulate the carbonaceous material within the gasifier **500**. However, any other suitable type of transporter can be used for the carbonaceous material. Alternatively, gravity can be used to feed carbonaceous material to the drying module **400**, particularly when the hopper geometry facilitates (e.g., funnels) carbonaceous material transportation. This variation can decrease the need for a driven material transporter, and can decrease the amount of energy required to run the gasifier **500**. The carbonaceous material within the hopper **200** is preferably pre-heated and partially dried by the hot air from the radiator blast of the engine **810**, but can alternatively be pre-heated and dried from heat radiating from the drying module **400**, heat radiating from the body of the gasifier **500**, or simply dried by allowing the moisture in the carbonaceous material to equilibrate with the ambient environment. The carbonaceous material in the hopper **200** can additionally be pre-heated/dried from any other heat source. In one variation, the carbonaceous material transporter **220** is angled relative to the skid **110** (e.g., relative to a normal vector or broad face of the skid **110**). This can be particularly desirable when the hopper base **210** is angled, wherein the angle of the carbonaceous material transporter **220** is preferably substantially the same as the hopper base **210** angle. The carbonaceous material transporter **220** preferably includes first end and a second end, wherein the first end is preferably lower than the second end.

The power generation system **10** can additionally include airlock **300** that functions as an airlock transition point that seals the gasifier **500**, preventing gas (air) ingress or egress, while maintaining the flow of material between the hopper **200** and the drying module **400**. However, some variations of the power generation system **10** an exclude an airlock, wherein the hopper preferably maintains an air seal for the remainder of the power generation system **10**. Effectively, the airlock **300** maintains an air seal in the body of the gasifier **500**, more preferably with the inlet of the drying module **400**,

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such that airflow within the body of the gasifier **500** can be controlled, which is desirable because pyrolysis in the pyrolysis module **510** must occur in the absence of oxygen and because achieving a strong air blast out the nozzles in the reaction module requires pulling or pushing against a sealed reactor environment. Furthermore, the airlock **300** feeds carbonaceous material from the ambient environment into the sealed system. The airlock **300** is preferably arranged within the hopper boot (e.g., hopper base **210**), but can alternatively be arranged between the hopper boot and the drying module **400**, within the hopper **200** before the hopper boot, or within the drying module **400**. Alternatively, the airlock **300** can be located below a raised end of the carbonaceous material transporter, such as when the carbonaceous material transporter is angled relative to the skid **110**. The airlock **300** arrangement allows for the body of the gasifier **500** to hold an air seal while allowing the hopper **200** to be open to the ambient atmosphere. This airlock configuration confers two benefits: first, airflow within the body of the gasifier **500** can be controlled. Second, the airlock placement allows the hopper **200** to be open to the ambient environment. This permits the hopper **200** to be collapsible, allowing for easy transportation (e.g., during shipping); eliminates the need for airtight installation of a new hopper, allowing the hopper volume to be easily changed; and allows for the carbonaceous material pre-drying (in the hopper **200**) to the ambient environment, which removes moisture from the system such that moisture introduction into the reactor module is reduced, thereby preventing thermal drag on reactor temperatures. The airlock **300** is preferably a rotary airlock feeder, but can alternatively be a single or double sluice valve, a double dump mechanism, an iris valve, a rotating plate valve, or any suitable airlock **300**. The rotary airlock feeder is preferably oriented with the rotational axis perpendicular to a gravity vector, but can alternatively be oriented with the rotational axis parallel to the gravity vector, or oriented in any suitable position. The carbonaceous material transport mechanism preferably feeds carbonaceous material to the airlock **300**, but the weight of the carbonaceous material can also feed the carbonaceous material to the airlock **300**.

The drying module **400** of the power generation system **10** functions to hold and dry a quantity of the carbonaceous material. The drying module **400** is preferably located within the air-sealed environment of the gasifier **500**, and preferably includes a portion arranged substantially below the hopper **200**, more preferably below the airlock **300**, such that undried carbonaceous material is dropped into the drying module **400**. However, the drying module **400** can be arranged beside the hopper **200** or in any other suitable position. The drying module **400** preferably includes a carbonaceous material inlet in communication with the hopper **200** that allows carbonaceous material into the drying module **400**, and a carbonaceous material outlet that allows carbonaceous material out of the drying module **400** and into the pyrolysis module **510**. Carbonaceous material is preferably transported through the airlock **300** from the hopper **200**, through the carbonaceous material inlet, and into the drying module **400**, wherein the carbonaceous material is preferably heated and dried by the gaseous output of the reactor module **550**. However, carbonaceous material can be transported through an open hole in the hopper bottom into the drying module **400**. As the carbonaceous material is heated, the water content of the carbonaceous material is vaporized. The water vapor will still pass through the reactor, but with substantially less thermal drag, given the energy intensive vaporization stage is not reliant on heat from combustion in the gasifier. The drying module **400** is preferably a traditional auger boot, with two

angled walls and two vertical walls, or a cylindrical vessel, but can be of any suitable geometry. The drying module **400** preferably fastens to the hopper **200** or airlock **300** through endplates (e.g., by nuts and bolts, screws, or any other suitable fastening mechanism), but can otherwise fasten to the hopper **200** or airlock **300**.

In one variation, the drying module **400** is a straight channel entirely located below the hopper **200**, oriented with a longitudinal axis parallel the longitudinal axis of the hopper **200**. The drying module **400** is preferably substantially smaller than the hopper **200**, and is preferably located near the center of the hopper **200** but can alternatively be eccentrically arranged relative to the hopper center. The inlet is preferably the top of the drying module **400** (the part proximal the hopper **200**), and the outlet is preferably arranged on the side of the drying module **400** near the bottom of the drying module **400** (the part distal the hopper **200**). The bulk of the drying module **400** is preferably offset from and arranged above the pyrolysis module **510** (e.g., distal the skid **110**), such that only the substantially dried carbonaceous material is pulled from the bottom of the drying module **400** and transported into the pyrolysis module **510**. Because water vapor flows upward and the wettest carbonaceous material is fed in from the top, the carbonaceous material at the bottom of the drying module **400** is the driest material, and the material that is most fit to undergo pyrolysis. In another variation, the drying module **400** includes a channel extending at an angle from the hopper **200** to the pyrolysis module **510**, wherein the inlet is preferably arranged below the hopper **200** (or airlock **300**) and the outlet is preferably connected to the top of the pyrolysis module **510**, distal the reactor module **550**. However, the inlet and outlet of the drying module **400** can be otherwise arranged.

The drying module **400** can additionally include a first heat exchanger **420** thermally coupled to the drying module **400**. The first heat exchanger **420** preferably includes an inner jacket that is formed by rolling a continuous piece of sheet metal and preferably includes flange rings coupled to the ends of the cylinder formed by the sheet metal and end plates fastened (e.g., using nuts and bolts or any other suitable fastener) to the sheet metal and flange rings that substantially maintains the rolled shape of the sheet metal. The drying module **400** can be defined within the lumen of the inner jacket, or can be a separate component located within the lumen defined by the inner jacket. The first heat exchanger **420** preferably also includes an outer jacket, formed using substantially similar materials and methods as the inner jacket, that couples around the inner jacket through the endplates (e.g., by nuts and bolts, a weld, or any other suitable fastening mechanism). Gaseous fuel from the reactor module **550** is preferably introduced into the first heat exchanger **420** to dry the carbonaceous material, but gaseous fuel from the particulate separator **600** or gaseous fuel from the third heat exchanger **560** can alternatively be used. The gaseous fuel path preferably extends from the particulate separator **600** or the reactor module **550** up to the first heat exchanger **420**. The inlet of the first heat exchanger **420** is preferably arranged proximal the outlet of the drying module **400**, but can alternatively be proximal the inlet of the drying module **400**. The outlet of the first heat exchanger **420** is preferably fluidly connected to the filtration module **700** or to the fuel inlet of the engine **810**. The temperature of the gaseous fuel that exits directly from the reduction reaction in typical gasifiers can be around 700-900° C., but the particulate separator **600** and/or the third heat exchanger **560**, which preheats the incoming air, can lower the temperature of this gaseous fuel output to around 200° C.-300° C. More preferably, the gaseous fuel

output temperature is preferably lowered to a range of around 150° C.-220° C. before ingressing into the first heat exchanger **420**, such that maximal drying can be achieved without inducing pyrolysis. Preventing pyrolysis in the drying section can be desirable to avoid the build up of tar condensates and generally maintain the cleanliness of the system. However, any other suitable temperature can be used.

To facilitate transport of the carbonaceous material into the pyrolysis module **510**, the drying module **400** can additionally include a second carbonaceous material transporter **440** that transports the dried carbonaceous material from the drying module **400** into the pyrolysis module **510**. The material transporter is preferably an auger, but can also be a drag chain, conveyor, plug feeder, walking floor or any other type of transporter. The second carbonaceous material transporter **440** preferably pulls carbonaceous material from the bottom of the drying module **400**, through the carbonaceous material outlet of the drying module **400**, and drops the dry carbonaceous material into the pyrolysis module **510**. Alternatively, the second carbonaceous material transporter **440** can function as a lift, transporting the carbonaceous material through the drying module **400** from a lower drying module inlet to a higher drying module outlet. The second carbonaceous material transporter **440** is preferably substantially similar to the first, and preferably operates at substantially the same speed, such that carbonaceous material flow through the system is substantially constant. However, the second carbonaceous material transporter **440** can be a substantially different mechanism, run at a substantially different speed, and/or operated by a substantially different method.

The gasifier **500** of the power generation system **10** functions to produce gaseous fuel from the carbonaceous material. The gasifier **500** is preferably substantially fluidly sealed, aside from the air input of the reactor module **550**, and is preferably capable of maintaining an internal pressure higher or lower than that of the ambient environment. The gasifier **500** is preferably fluidly connected to the fuel input of the engine module **800**, wherein the engine module **800** converts the gaseous fuel into rotary motion, then to electrical power. The gasifier **500** preferably includes the pyrolysis module **510** and the reactor module **550**. The gasifier **500** preferably additionally includes a second and third heat exchanger **560** that uses waste heat from the generated gaseous fuel and the engine exhaust to heat the air heat exchanger **560**, drying module **400**, and pyrolysis module **510**, respectively. The gasifier **500** can additionally include any other suitable generator component.

The pyrolysis module **510** of the gasifier **500** functions to pyrolyze dry carbonaceous material into tar gas and charcoal. More specifically, the carbonaceous material is heated at a substantially high temperature (typically above 220° C.), in the substantial absence of oxygen, thus distilling the carbonaceous material into tar gas and charcoal. Tar gas and charcoal are the precursors necessary for the following combustion and reduction stages in gasification. The pyrolysis module **510** preferably includes a dried carbonaceous material inlet that allows dried carbonaceous material from the drying module **400** to enter the pyrolysis module **510** and a tar gas and charcoal outlet that allows the tar gas and charcoal to exit the pyrolysis module **510**, preferably into the reactor module **550**. Similar to the drying module **400**, the pyrolysis module **510** preferably also of a generally cylindrical structure that includes an inner jacket and an outer jacket that cooperatively define an annular space in between the inner jacket and the outer jacket. The pyrolysis module **510** is preferably constructed using similar or identical materials and methods as

described above for the drying module **400**, but can alternatively be constructed using any other suitable material and method.

The annular space surrounding the pyrolysis module **510** preferably defines the second heat exchanger **520**. Exhaust gas from the engine module **800** is preferably piped into the second heat exchanger **520** to pyrolyze the dry carbonaceous material within the pyrolysis module **510**. The temperature of the exhaust gas that enters this annular space can be in the range of 500° C.-700° C., and preferably contains enough heat energy to heat the dried carbonaceous material to temperatures that pyrolyze the dried carbonaceous material into tar gas and charcoal. Temperatures necessary to pyrolyze carbonaceous material are generally above 220° C. The second heat exchanger **520** preferably includes an inlet fluidly connected to the exhaust of the engine **810** of the engine module **800**, and preferably includes an outlet fluidly connected to the ambient environment but alternatively connected to any other suitable fluid reservoir. To minimize the connection length between the pyrolysis module **510** and the engine exhaust, as well as to minimize thermal losses from the exhaust gas stream, the pyrolysis module **510** is preferably arranged adjacent the engine exhaust, more preferably in a corner adjacent the engine **810** of the engine module **800** but alternatively in any other suitable location. The exhaust path length is preferably substantially less than a side of the skid **110**, preferably less than the length of enclosure, and more preferably less than the width of the enclosure. The fluid connection from the engine **810** to the pyrolysis module **510** is also preferably insulated to minimize thermal loss from the transported exhaust gasses, but can similarly be uninsulated.

The pyrolysis module **510** is preferably located directly above the reactor module **550**, and is preferably coupled to the reactor module **550** through flange rings and endplates. Furthermore, the pyrolysis module **510** is preferably located in a corner of the square footprint. However, any other suitable method of arranging the pyrolysis module **510** can be used. The tar gas and charcoal outlet of the pyrolysis module **510** preferably extends into the reactor module **550**, allowing tar gas and charcoal to exit the pyrolysis module **510** and flow directly into the reactor module **550** through gravity. The sealed reactor environment will cause the tar gas to propagate into the reactor module **550**, and char will similarly enter the reactor module via gravity. However, the tar gas and charcoal can be actively moved into the reactor module **550**, for example with a motor, or can be transported into the reactor module **550** using any suitable method. While the tar gas and charcoal outlet of the pyrolysis module **510** can extend into the reactor module **550**, the heat from the reactor module **550** preferably does not substantially affect heat within the pyrolysis module **510**, thus decreasing the thermal load on the reactor module **550**. The depth that the tar gas and charcoal outlet extends into the reactor is preferably where the temperature of the pyrolysis module **510** is substantially higher than that of the reactor module **550** at that particular level, discouraging heat transfer from the reactor module **550** into the pyrolysis module **510**. However, any other suitable arrangement of the pyrolysis module **510** can be used.

The reactor module **550** of the gasifier **500** functions to combust the tar gas and to reduce the combusted tar gas with the charcoal. More specifically, the tar gas is placed in contact with oxygen at combustion temperatures to combust, and the combusted tar gas is placed in contact with the charcoal to reduce into carbon monoxide (CO) and hydrogen (H₂), which can then be used as gaseous fuel. As described above, the tar gas and charcoal enters the reactor module **550** from the tar gas and charcoal outlet of the pyrolysis module **510**. The

reactor module **550** preferably includes an air path that channels air with a suitable amount of oxygen from any suitable air source (e.g., ambient air) to the interior of the reactor module **550**, allowing combustion and reduction of the tar gas. The reactor module **550** is preferably of the Imbert reactor type (e.g., the throat and nozzles), but can alternatively be any other suitable reactor type.

The reactor module **550** is preferably located inside a substantially cylindrical gas cowl that substantially envelops the reactor module **550**. The gas cowl also preferably includes sheet metal, flange rings, and end plates that maintain the shape and fasten the gas cowl to the reactor module **550**. The gas cowl preferably accommodates a variety of reactor types. The gas cowl and the reactor module **550** preferably cooperatively define a third heat exchanger **560**, an annular space between the reactor module **550** and the gas cowl. The gaseous fuel outlet of the reactor module **550** is preferably fluidly coupled to the gas cowl. The gaseous fuel outlet is preferably located at the bottom of the reactor module **550** (e.g., proximal the skid **110** or pallet, distal the pyrolysis module **510**), such that the gaseous fuel exits after the reduction reaction from the bottom of the reactor and is directed upwards and around the reactor module **550** by the gas cowl. However, the gaseous fuel outlet can be located at any suitable position. This hot, gaseous fuel is preferably used to preheat the air used to combust the tar gas within the reactor module **550**, wherein the air path is preferably located within the annular space between the reactor module **550** and the gas cowl (third heat exchanger **560**). The air path preferably wraps around all or a portion of the reactor module perimeter prior to entering the reactor module interior. The air path inlet is preferably proximal the reactor module bottom (e.g., end of reactor module **550** proximal the skid **110**) and the air path outlet is preferably proximal the reactor module **550** top (e.g., end of reactor module **550** distal the skid **110** and proximal the pyrolysis module **510**), allowing the air path to traverse substantially the entire length of the third heat exchanger **560**. However, the air path inlet and outlet can be located in any suitable position. This air path arrangement substantially increases the output temperature of the air introduced into the reactor module **550** and substantially decreases the output temperature of the gaseous fuel. Additionally, as a result of the heat transfer, the temperature of the reactor module **550** is substantially lower at the top than at the bottom, allowing the pyrolysis module **510** to be inserted into the reactor module **550** without substantial heat transfer from the reactor module **550** to the pyrolysis module **510** and facilitating transport of tar gas and charcoal into the reactor module **550**. However, any other arrangement suitable arrangement of the reactor module **550** can be used.

The gas cowl of the third heat exchanger **560** can also include a sediment collector located below the reactor module **550** that collects particulates that are extracted from the gaseous fuel and/or the ash that results from the charcoal after the combusted tar gas is reduced with the charcoal. The sediment collector is preferably removable to allow for the sediments to be disposed of after use. However, any other suitable arrangement of the sediment collector may be used.

The particulate separator **600** of the power generation system **10** functions to extract particulates from the gaseous fuel prior to fuel ingress into the first heat exchanger **420**. The particulate separator **600** can additionally extract heat from the gaseous fuel output prior to using the heat to dry the carbonaceous material. The particulate separator **600** is preferably fluidly connected to the outlet of the third heat exchanger **560** and to the inlet of the first heat exchanger **420**. The particulate separator **600** preferably includes a gaseous

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fuel input that receives gaseous fuel from the gaseous fuel outlet of the gas cowling and a gaseous fuel output that is coupled to the gaseous fuel inlet of the first heat exchanger 420. The particulate separator 600 is preferably operable at high temperatures. The particulate separator 600 can be a cyclone module that uses centrifugal force to separate particulates from the gaseous fuel, but can alternatively include any other suitable particulate separator. In the process of routing the gaseous fuel through the cyclone module, heat can additionally be radiated out into the ambient environment. To this end, the particulate separator 600 can additionally include heat-radiating features (for example, heat fins, a pin matrix, etc.) that facilitate the extraction of heat from the gaseous fuel output during particulate extraction. Furthermore, as heat loss from the generated fuel is inversely proportional to the fuel flow rate out of the reactor module (e.g., more heat is lost when fuel flows at lower rates), cyclone operation can be optimized to adjust the amount of heat extracted by the cyclone (e.g., by adjusting the fuel flow rate within the cyclone, as determined empirically or from a chart). The particulate separator 600 is preferably arranged underneath the first heat exchanger 420 to minimize connection length, but can similarly be mounted elsewhere on the gas cowling. More preferably, the particulate separator 600 is fluidly connected to a portion of the first heat exchanger 420 proximal the skid 110. Alternatively, the particulate separator 600 can be thermally connected to a water path that receives water from a water source, such as filter condensate, hopper condensate, or water from a dedicated external water source. The heat extracted by the particulate separator 600 preferably heats the water within the water path, vaporizing the water. The water vapor (steam) can subsequently be fed into the reactor module to improve gas quality. However, any other suitable arrangement of the particulate separator 600 can be used. The particulate separator 600 can be arranged adjacent the hopper 200 to dry the carbonaceous material with the extracted heat. The portion of the particulate separator 600 proximal the hopper 200 preferably has a larger surface area than portions of the particulate separator 600 distal the hopper 200, but can alternatively have a similar surface area or any other suitable surface area. In one variation of the power generation system 10, the particulate separator 600 is located below the hopper 200, between the hopper 200 and the skid 110. In another variation of the power generation system 10, the particulate separator 600 is located adjacent the hopper 200, wherein both the hopper 200 and the particulate separator 600 are mounted to or defined by the skid 110. However, the particulate separator 600 can be otherwise arranged.

The filtration module 700 of the power generation system 10 functions to extract fine particulates and any remaining tar gas from the gaseous fuel, such that the fuel provided to the carburetor of the engine 810 is clean, engine-grade fuel. The filtration module 700 is preferably a packed bed, made from a cylindrical steel drum that is filled with a filtration media (e.g., fine grain wood chips, activated carbon, etc.). Alternatively, the filtration module 700 can be a water scrubber, electrostatic precipitator, ceramic candle, bag house, or any other suitable filtration system. The filtration module 700 preferably includes a gaseous fuel inlet, located substantially near the bottom of the filtration module 700 (distal from the drying module 400), and a gaseous fuel outlet, located substantially near the top of the filtration module 700. The gaseous fuel inlet is preferably fluidly connected to the first heat exchanger 420, and the gaseous fuel outlet is preferably fluidly connected to an engine fuel inlet. The gaseous fuel is preferably piped from the first heat exchanger 420 (the annular space of the drying module 400), after it has dried the

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carbonaceous material, into the bottom of the filtration module 700 through the gaseous fuel inlet, wherein the gaseous fuel is filtered as it rises through the filtration material, emerging from the gaseous fuel outlet as substantially clean gaseous fuel. The filtered fuel is then preferably provided to the carburetor of the engine module 800. The filtration module 700 is preferably located adjacent to the drying module 400 to minimize the connection length, and is also preferably located adjacent the engine fuel inlet and the pyrolysis-reactor assembly. In one variation of the power generation system 10, the filtration module 700 can be located in a corner adjacent the pyrolysis-reactor assembly corner in the square footprint. In another variation of the power generation system 10, the filtration module 700 can be located along a longitudinal edge of the enclosure, adjacent the pyrolysis-reactor assembly, more preferably between the pyrolysis-reactor assembly and the hopper 200. However, the filtration module 700 can be located in any suitable position. The power generation system 10 preferably includes one filtration module 700, but can alternatively include any suitable number of filtration modules. Multiple filtration modules preferably have substantially similar configurations, but can alternatively have different configurations. Multiple filtration modules are preferably fluidly coupled in series, but can alternatively be fluidly coupled in parallel.

The power generation system 10 can require a startup system that pulls gas through the system. To this end, the power generation system 10 can additionally include an ejector module, blower, fan or other gas drive mechanism coupled to the filtration system. The ejector module is preferably coupled to the filtration system substantially near the gaseous fuel outlet, but can alternatively be otherwise coupled to the filtration system, the drying module 400, or any other connection point downstream from the reactor module 550. The ejector module output is preferably coupled to a combustion flare, with variable air mixing system to ensure full combustion. The ejector system is preferably coupled to the top of the filtration system, more preferably to the center of the cap, wherein the gas flare preferably extends horizontally out from the ejector system into the corner of the square footprint, and extends vertically through the corner space provided by the circular hopper of the drying module 400. The ejector system is coupled to an air input, compressed air is preferably pumped into the ejector system through the air input. When the high velocity of air suddenly expands into the wide ejector system, it entrains a large quantity of gas, producing a negative pressure that pulls gas through the filtration module 700 and, subsequently, the rest of the power generation system 10. However, the power generation system 10 can additionally include a fan at the air inlet of the reactor module that pushes air into the reactor module, a fan or vacuum-generating mechanism (e.g., the engine) that pulls gaseous fuel out of the reactor module, or any other suitable mechanism that induces fluid movement.

The engine module 800 of the power generation system 10 functions to convert gaseous fuel into electricity. The engine module 800 (genset) preferably includes an internal combustion engine 810 that converts the gaseous fuel into motion, an alternator 850 that converts the motion into electricity, and a drive mechanism that transfers motion from the engine 810 to the alternator 850. Gaseous fuel is preferably introduced into the fuel inlet of the engine, more preferably into the carburetor of the engine from the filtration module 700, but can alternately be introduced from the first heat exchanger 420, the heat exchanger, or directly from the third heat exchanger 560/reactor module 550. As aforementioned, the engine 810 and alternator 850 are preferably arranged in an in-line con-

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figuration, but can alternatively have any suitable configuration. In comparison to a side-by-side engine-alternator configuration, the in-line configuration allows for a shorter drive mechanism to be used, and allows for the other components of the power generation system 10 to fit within the enclosure footprint. In one variation of the power generation system 10, the engine 810 occupies one corner of the enclosure footprint and the alternator 850 occupies an adjacent corner. In another variation of the power generation system 10, the alternator 850 occupies a corner of the enclosure footprint and the engine 810 is located along the enclosure body (e.g., along a longitudinal edge but alternatively along a lateral edge). The engine 810 preferably includes a radiator 820 and an exhaust. The radiator 820 is preferably arranged adjacent and/or below the hopper 200, but can alternatively be arranged elsewhere. The radiator 820 can include a plenum that directs and/or contains heated air radiated from the engine. The radiator position is preferably adjustable, but can alternatively be static. The radiator plenum volume and/or direction is preferably adjustable, but can alternatively be statically fixed. In one variation of the power generation system 10, the radiator 820 is arranged below a raised end of the hopper base 210. The engine 810 is preferably oriented with the engine exhaust proximal the pyrolysis module 510, but can alternatively be oriented with the engine exhaust distal the pyrolysis module 510. The piping leading from the engine exhaust to the pyrolysis module 510 preferably passes below the hopper 200, but can alternatively pass adjacent the hopper 200 or be distal from the hopper 200. The alternator 850 is preferably oriented with a power interface (e.g., plugs) proximal an edge of the enclosure 100, but can alternatively be arranged in any suitable manner. The engine 810 is preferably an internal combustion engine, and is preferably an inline cylinder configuration (e.g., 2, 3, 4, 5, or 6 cylinders, etc.) so as to optimally locate the exhaust manifold output immediately adjacent to the pyrolysis module input. However, any engine configuration is acceptable, with appropriate piping solutions to distribute the hot exhaust to the pyrolysis module. The alternator 850 is preferably a standard 10 kW alternator, but can alternatively be a 20 kW alternator, a 100 kW alternator, or any other suitable alternator that converts motion from the engine 810 into electricity. The drive mechanism is preferably a direct drive inline engine-alternator coupled solution, however a belt drive that couples the crankshaft of the engine to the rotor of the alternator 850 or any other suitable drive mechanism can alternatively be used.

2. First Specific Example

Configuration for a Standard Shipping

In a first specific example of the power generation system 10, as shown in FIGS. 1A, 1B, 3A, and 3B, the power generation system 10 is configured to fit on a standard shipping pallet (e.g., an ISO standard pallet, North American standard pallet as shown in FIGS. 2A and 2B, European pallet, Australian pallet, etc.). The shipping pallet preferably forms the skid 110. The enclosure can additionally include enclosure walls 120 that are preferably perpendicularly mounted to the skid 110 to form a fully enclosed system. The enclosure can additionally include a top 130. The pallet preferably has a broad face defining a first corner adjacent a second corner, a third corner adjacent the second corner, and a fourth corner adjacent the first and third corners, wherein the four corners are preferably defined in a common plane. The power generation system 10 includes a reactor module 550 located in the first corner, a pyrolysis module 510 coaxially arranged above

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the reactor module 550 (e.g., such that the reactor module 550 is between the pyrolysis module 510 and the pallet), a drying module 400 located offset from and above the pyrolysis module 510 (e.g., such that the pyrolysis module 510 is between the drying module 400 and the pallet), and an engine module 800 (genset) occupying the second and third corners. The power generation system 10 can optionally include an airlock 300 located above the drying module 400 (e.g., the drying module 400 is located between the airlock 300 and the pallet) that creates a substantially fluid impermeable seal with the drying module 400. The fluid path between the drying module 400, pyrolysis module 510, reactor module 550, and engine module 800 is preferably substantially fluidly sealed from the ambient environment, sans the air input for the reactor module 550. The drying module 400 can vertically overlap with some of the generator components (e.g., the bottom of the drying module 400 overlaps with the top of the pyrolysis module 510), but is preferably arranged above the rest of the generator components, substantially near the center of the pallet footprint. The drying module 400 can additionally include a material transporter located proximal the pallet within the drying module 400, wherein the material transporter moves carbonaceous material from the bottom of the drying module 400 into the pyrolysis module 510. The engine module 800 and reactor module 550 are preferably directly mounted to the pallet, while the pyrolysis module 510 is preferably mounted to the reactor module 550 and the drying module 400 is preferably mounted to the pyrolysis module 510. The engine 810 preferably occupies the second corner while the alternator 850 occupies the third, but the engine module 800 can be otherwise arranged. The power generation system 10 can additionally include a hopper 200 above the drying module 400, such that the drying module 400 is between the hopper 200 and the pallet. The hopper 200 preferably has a major dimension (e.g., diameter) substantially equivalent to the width or length of the pallet. The hopper 200 is preferably an open cylinder with a closed end (wherein the closed end is proximal the pallet), but can alternatively be a cone with the apex proximal the pallet or have any other suitable configuration. The hopper 200 is preferably collapsible and comes pre-mounted to the gasifier 500, but can alternatively be shipped as a separate component (in a rigid or collapsed state) or be rigid and pre-mounted to the gasifier 500. The hopper 200 is preferably porous (e.g., made of mesh), but can alternatively be substantially solid. The hopper 200 preferably forms an air seal with the rest of the power generation system 10, but can alternatively be open to the ambient environment and include an airlock 300. The airlock 300, when included, is preferably located in the bottom of the hopper 200 (the portion of the hopper 200 proximal the pallet), but can alternatively be located in the drying portion or between the two components. The airlock 300 is preferably a rotary airlock 300, and is preferably oriented with the rotational axis substantially parallel to the broad face of the pallet. The hopper 200 can additionally include a material transporter that moves carbonaceous material from the hopper base 210 to the airlock 300. The hopper material transporter preferably traces the profile of the hopper base 210, but can alternatively be at an angle relative to the hopper base 210. The power generation system 10 can additionally include a filtration module 700 located in the fourth corner, wherein the filtration system is preferably mounted to the pallet. However, the pyrolysis-reactor assembly and filtration module corners can be switched. The power generation system 10 can additionally include a particulate separator 600 located between the filtration module 700 and the alternator 850, near the center of the pallet footprint. The particulate separator 600

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can include a cyclone module and heat-radiating features, such as fins, but can alternatively be any suitable particulate separator **600**. The particulate separator **600** is preferably located below the drying module **400** (e.g., such that the particulate separator **600** is located between the drying module **400** and the pallet). The particulate separator **600** can be directly below the drying module **400** (e.g., share a common vector normal to the pallet broad face), offset from the drying module **400**, located below a portion of the hopper **200**, or arranged in any other suitable location. The particulate separator **600** is preferably mounted to the reactor module **550** (e.g., about the reactor module outlet), but can alternatively be mounted to any suitable generator component. This arrangement preferably additionally minimizes the amount and/or length of interconnects by reducing the length of the plumbing runs and arranging the output exhaust pipe of the engine **810** is preferably arranged proximal to the pyrolysis module inlet. Furthermore, the electrical controls and instrumentation are preferably arranged on a single conduit **20** (e.g., wiring conduit including electrical wiring and/or fluid channeling) extending along the center of shipping pallet, in between the reactor module **550** and the engine module **810**. Power and signal connections from the gasifier and engine modules are preferably designed to face inward, toward the single distribution conduit **20**, for easy connection and minimal wiring. However, the power generation system **10** can include any suitable number of conduits. The conduit preferably extends through the center of the system, wherein the conduit preferably extends parallel to a broad face of the pallet, but can alternatively extend perpendicular to the broad face of the pallet (e.g., vertically). The conduit preferably additionally includes a user control module **25** (e.g., arranged along or proximal an edge between the filtration module **700** and the engine module **800**), but can alternatively include any other suitable user input interface. This arrangement preferably additionally enables single-side operation by arranging all the electrical controls and plumbing valves along a single side.

The power generation system **10** preferably includes a first heat exchanger **420** fluidly coupled to the third heat exchanger **560** (and thus, the outlet of the reactor module **550**) and thermally coupled to the drying module **400**, wherein the first heat exchanger **420** heats the carbonaceous material with waste heat from the gaseous fuel. More preferably, the inlet of the first heat exchanger **420** is fluidly coupled to the particulate separator **600**. The outlet of the first heat exchanger **420** is preferably fluidly connected to the filtration module **700** but can alternatively be directly connected to the carburetor of the engine. The first heat exchanger **420** is preferably an annular channel surrounding the drying module **400**, but can be any suitable heat exchanger. The power generation system **10** preferably includes a second heat exchanger **520** fluidly coupled to the engine exhaust and thermally coupled to the pyrolysis module **510**, wherein the second heat exchanger **520** heats the dried carbonaceous material with waste heat from the exhaust gas. The outlet of the second heat exchanger **520** is preferably coupled to the ambient environment. The second heat exchanger **520** is preferably an annular channel surrounding the pyrolysis module **510**, but can be any suitable heat exchanger. The power generation system **10** preferably includes a third heat exchanger **560** fluidly coupled to the outlet of the reactor module **550** and thermally coupled to an air intake for the reactor module **550**, wherein the third heat exchanger **560** heats the air used for combustion. More preferably, the third heat exchanger **560** receives hot gaseous fuel from the reactor module outlet and egresses gaseous fuel into the particulate separator **600**. The third heat exchanger **560** is

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preferably an annular channel surrounding the reactor module **550**, but can be any suitable heat exchanger.

However, the power generation system components can be otherwise arranged within the pallet footprint.

3. Second Specific Example

Configuration for a Shipping Container Enclosure

In a second specific example of the power generation system **10**, as shown in FIGS. **4A** and **4B**, the power generation system **10** is configured to fit within a standard intermodal shipping container enclosure. The power generation system **10** can be arranged within a 20 foot intermodal container, a 40 foot intermodal container, a 40 foot high cube container, a 45 foot high cube container, or any other suitable shipping container. The shipping container enclosure preferably includes a rectangular skid **110**, four walls extending perpendicularly from the skid **110**, and a top **130** statically connected to the ends of the four walls distal the skid **110**. The shipping container enclosure preferably has a first and a second end perpendicular the enclosure longitudinal axis (e.g., the short sides of the shipping enclosure). The enclosure is preferably divided along its length (longitudinal axis) into a enclosure side occupying the first end and a hopper side occupying the second end, wherein the hopper **200** is located in the hopper side and preferably occupies the entire second end of the enclosure. The hopper **200** preferably occupies substantially half of the enclosure, but can alternatively occupy more or less of the enclosure. The hopper **200** is preferably cooperatively defined by an auxiliary wall and three adjacent walls of the enclosure, and can additionally be defined by the top and/or base. The hopper **200** is preferably an open hopper (e.g., open to ambient environment), but can be any suitable hopper. The hopper **200** preferably has an angled bottom relative to the skid **110**, wherein the hopper base **210** preferably has a first end and a second end. The first end of the hopper base **210** is preferably the short edge of the skid **110**, and the second end of the hopper is preferably proximal the center of the skid **110**, wherein the distance between the hopper base **210** and the skid **110** preferably increases with proximity to the center of the skid no (e.g., the second end of the hopper base **210** is further away from the skid **110** than the first end of the hopper base **210**). The hopper base **210** is preferably substantially planar, but can alternatively be curved or have any suitable geometry. The hopper base **210** preferably meets the skid **110** along the short edge of the second end of the enclosure, but can alternatively meet the skid **110** along a longitudinal wall. The hopper **200** preferably additionally includes a material transporter that functions to move carbonaceous material from the hopper base **210** toward the enclosure center. The material transporter preferably moves carbonaceous material in a vector substantially parallel to the hopper base **210**, but can alternatively move carbonaceous material in a vector at an angle to the hopper base **210**. The hopper base **210** is preferably porous (e.g., perforated, formed from a mesh, etc.), but can alternatively be solid. The walls and/or top **130** defining the hopper base **210** are preferably solid, but can alternatively be porous or articulate relative to the skid **110** (e.g., have doors). The hopper base **210** and auxiliary wall are preferably substantially thermally conductive and made of metal, but can alternatively be made of any suitable material.

The remainder of the generator components are preferably located in the component side, wherein the first end includes a first and a second corner. The pyrolysis module **510** and reactor module **550** are preferably located in the first corner,

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wherein the pyrolysis module **510** is preferably coaxially arranged above the reactor module **550** (e.g., such that the reactor module **550** is located between the pyrolysis module **510** and the skid **110**). The reactor module **550** is preferably mounted to the skid **110**, while the pyrolysis module **510** is preferably mounted to the reactor module **550** end distal the skid **110** (opposing end). The engine module **800** is preferably located in the second corner, arranged with the engine **810**, drive mechanism, or alternator longitudinal axis parallel to the footprint longitudinal axis. The engine module **800** is preferably arranged with the alternator **850** in the second corner and the engine module **800** between the hopper **200** and the alternator **850**. The engine module **800** is preferably arranged with a control panel and/or power interface proximal a longitudinal edge of the enclosure, but can alternatively be arranged with the control panel and/or power interface proximal a lateral edge of the enclosure. The engine module **800** can be arranged with the engine radiator proximal the hopper auxiliary wall, wherein the radiator blast can ingress into the hopper and pre-heat and pre-dry the carbonaceous material within the hopper **200**. The radiator blast is preferably directed into the hopper **200**, but the hot air from the radiator **820** can alternatively be fed to the hopper **200** through a pipe or any suitable mechanism. However, the engine module **800** can be otherwise arranged within the enclosure **100**. The engine module **800** and the alternator **850** are preferably mounted directly to the skid **110**. The filtration modules are preferably located along a longitudinal edge shared with the pyrolysis/reactor modules **550**, between the gasifier **500** and the hopper **200**. The filtration modules are preferably mounted directly to the skid **110**. The drying module **400** preferably extends from the bottom of the hopper **200** to the top of the pyrolysis module **510** along a longitudinal portion of the enclosure. The drying module **400** preferably has a first and a second end, wherein the first end of the drying module **400** is preferably located below the second end of the hopper base **210** (e.g., between the second end of the hopper base **210** and the skid **110**), and the second end of the drying module **400** is preferably fluidly connected to the pyrolysis module **510** distal the skid **110**. The drying module **400** preferably includes a channel encapsulating a material transporter, more preferably includes a channel surrounding a heated auger. The drying module **400** can be mounted to the hopper base **210**, hopper auxiliary wall, or airlock **300** casing at the first end, and mounted to the pyrolysis inlet at the second end. The airlock **300** is preferably located between the second end of the hopper base **210** and the first end of the drying module **400**, and preferably forms a substantially fluid impermeable seal with the drying module **400**. The airlock **300** is preferably a rotary airlock **300** and oriented with the rotational axis substantially parallel the skid broad face, but any other type of air lock mechanism can be used. The end material transporter of the hopper **200** preferably extends over the airlock-hopper junction, such that carbonaceous material is dropped into the airlock **300**. The airlock **300** is preferably mounted to the hopper auxiliary wall and hopper base **210**. The particulate separator **600** can be located proximal the center of the enclosure **100**. Alternatively, the particulate separator **600** can be located between the engine **810** and the auxiliary wall of the hopper **200**, wherein the heat exchanger preferably heats and/or dries the carbonaceous material within the hopper **200** with waste heat extracted from the gaseous fuel. The outlet of the particulate separator **600**, more preferably the entirety of the particulate separator **600**, is preferably located below the inlet of the first heat exchanger **420** (e.g., between the first heat exchanger inlet and the skid **110**), but can alternatively be located elsewhere. The particu-

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late separator **600** is preferably mounted to the outlet of the third heat exchanger **560**, but can alternatively be mounted to any suitable component of the power generation system **10**. The enclosure **100** preferably has doors along the walls defining the enclosure first end, such that the generator components can be accessed.

The power generation system **10** preferably includes a first heat exchanger **420** fluidly coupled to the third heat exchanger **560** (and thus, the outlet of the reactor module **550**) and thermally coupled to the drying module **400**, wherein the first heat exchanger **420** heats the carbonaceous material with waste heat from the gaseous fuel. More preferably, the inlet of the first heat exchanger **420** is fluidly coupled to the particulate separator **600**. The outlet of the first heat exchanger **420** is preferably fluidly connected to the filtration module **700** but can alternatively be directly connected to the carburetor of the engine. The first heat exchanger **420** is preferably an annular channel surrounding the drying module **400** (e.g., extending along the longitudinal axis of the drying module **400**), but can be any suitable heat exchanger. The inlet of the first heat exchanger **420** is preferably proximal to the first end of the drying module **400** and the outlet of the first heat exchanger **420** proximal the second end of the drying module **400**. Alternatively, the outlet of the first heat exchanger **420** is proximal to the first end of the drying module **400** and the inlet of the first heat exchanger **420** is proximal the second end of the drying module **400**. Alternatively, the first heat exchanger **420** includes two fluid channels with opposing flow directions that are fluidly connected at one end, wherein the inlet and outlet of the first heat exchanger **420** can be located proximal the pyrolysis module **510** or the hopper **200**. The power generation system **10** preferably includes a second heat exchanger **520** fluidly coupled to the engine exhaust and thermally coupled to the pyrolysis module **510**, wherein the second heat exchanger **520** heats the dried carbonaceous material with waste heat from the exhaust gas. The outlet of the second heat exchanger **520** is preferably coupled to the ambient environment. The second heat exchanger **520** is preferably an annular channel surrounding the pyrolysis module **510**, but can be any suitable heat exchanger. The power generation system **10** preferably includes a third heat exchanger **560** fluidly coupled to the outlet of the reactor module **550** and thermally coupled to an air intake for the reactor module **550**, wherein the third heat exchanger **560** heats the air used for combustion. More preferably, the third heat exchanger **560** receives hot gaseous fuel from the reactor module outlet and egresses gaseous fuel into the particulate separator **600**. The third heat exchanger **560** is preferably an annular channel surrounding the reactor module **550**, but can be any suitable heat exchanger.

However, the generator components can be otherwise arranged in a standardized shipping container enclosure. Alternatively, any suitable arrangement of the modules can be used.

As a person skilled in the art will recognize from the previous detailed description and from the figures and claims, modifications and changes can be made to the preferred embodiments of the invention without departing from the scope of this invention defined in the following claims.

We claim:

1. A gasification based power generation system that converts carbonaceous material into electrical power, comprising:

a shipping pallet having a first corner adjacent a second corner, a third corner adjacent the second corner, and a fourth corner adjacent the third and first corners;
a hopper;

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a drying module arranged between the pallet and the hopper;
 a pyrolysis module arranged between the drying module and the pallet, the pyrolysis module offset from the drying module;
 a reactor module, located in the first corner, coaxially aligned with the pyrolysis module and arranged between the pyrolysis module and the pallet;
 an in-line alternator and engine system located in the second and third corners;
 a first heat exchanger fluidly coupled to an outlet of the reactor module and thermally coupled to the drying module;
 a second heat exchanger fluidly coupled an exhaust of the engine and thermally coupled to the pyrolysis module;
 a third heat exchanger fluidly coupling the outlet of the reactor module to the first heat exchanger, the third heat exchanger thermally connected to an air inlet for the reactor module.

2. The system of claim 1, further comprising a filtration system located in the fourth corner, wherein the first heat exchanger fluidly couples the outlet of the reactor module to an inlet of the filtration system, wherein an outlet of the filtration system is fluidly coupled to the inlet of the engine.

3. The system of claim 1, further comprising a heat extraction module fluidly connected between the third heat exchanger and the first heat exchanger, wherein the heat extraction module is located between the filtration system and the alternator.

4. The system of claim 1, wherein the hopper comprises an open-air hopper, the system further comprising an airlock connecting the hopper to the drying module, the airlock configured to form a substantially fluid impermeable seal with the drying module.

5. The system of claim 4, wherein herein the airlock is located in a bottom of the hopper.

6. The system of claim 5, wherein the hopper comprises a cylindrical structure having a substantially flat bottom arranged proximal and substantially parallel to a broad face of the pallet, the hopper further comprising a material transporter configured to move carbonaceous material to the airlock.

7. The system of claim 5, wherein the airlock comprises a rotary airlock.

8. The system of claim 7, wherein the airlock is oriented with a rotational axis parallel to a broad face of the shipping pallet.

9. The system of claim 5, wherein the hopper is collapsible.

10. The system of claim 1, wherein further comprising an enclosure having a height, wherein the shipping pallet forms a bottom of the enclosure, wherein a combined height of the reactor module, pyrolysis module, and drying module is less than an enclosure height.

11. The system of claim 1, further comprising a wiring conduit substantially centered between the first, second, third, and fourth corners, the wiring conduit extending in parallel to a broad face of the pallet, the wiring conduit electrically connected to power and signal connectors of the pyrolysis module, reactor module, and engine.

12. The system of claim 11, further comprising a control panel electrically connected to the wiring conduit, the control panel arranged along an edge connecting the third and fourth corners of the pallet.

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13. A gasification based power generation system that converts carbonaceous material into electrical power, comprising:

an enclosure having a first and a second end perpendicular to a longitudinal axis of the enclosure, the first end having a first and a second corner;

an open-air hopper arranged along the second end;

an airlock connected to an outlet of the hopper;

a drying module connected to the airlock, the airlock configured to form a substantially fluid impermeable seal with the drying module;

a gasifier comprising: a pyrolysis module arranged coaxially above a reactor module adjacent in the first corner, the pyrolysis module connected to the drying module;

an engine module comprising an in-line alternator and engine located in the second corner, the engine further comprising a radiator located between the engine module and the hopper;

a first heat exchanger fluidly connected to an outlet of the reactor module and thermally connected to the drying module;

a second heat exchanger fluidly connected to an outlet of the engine and thermally connected to the pyrolysis module;

a third heat exchanger fluidly connected between the outlet of the reactor module and the first heat exchanger, the third heat exchanger thermally connected to an air inlet of the reactor module.

14. The system of claim 13, wherein the heat exchange module is thermally coupled to the hopper.

15. The system of claim 14, wherein the hopper comprises a louvered floor.

16. The system of claim 13, wherein the first heat exchanger comprises an annular channel surrounding the drying module.

17. The system of claim 16, wherein the drying module comprises a channel and a material transporter extending therein.

18. The system of claim 13, further comprising a filtration module arranged between the gasifier and the hopper within the enclosure.

19. The system of claim 18, wherein the first heat exchanger fluidly connects the outlet of the reactor module to an inlet of the filtration module and an outlet of the filtration module is fluidly connected to an inlet of the engine.

20. The system of claim 13, wherein the enclosure comprises a standard intermodal shipping container.

21. The system of claim 20, wherein the standard shipping enclosure is a standard twenty-foot container.

22. The system of claim 13, wherein the hopper comprises a material transporter arranged along a hopper bottom.

23. The system of claim 22, wherein the material transporter is arranged at an angle to a normal axis of a bottom of the enclosure, the material transporter comprising a first end and a second end, the second end arranged distal the bottom of the enclosure across the air lock.

24. The system of claim 23, wherein the drying module comprises a first end and a second end, wherein the first end of the drying module is arranged between the airlock and the bottom of the enclosure, and the second end is connected to the pyrolysis module distal the reactor module.

25. The system of claim 24, wherein the airlock comprises a rotary airlock with a rotational axis parallel to the bottom of the enclosure.

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UNITED STATES PATENT AND TRADEMARK OFFICE
CERTIFICATE OF CORRECTION

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INVENTOR(S) : James Mason et al.

Page 1 of 1

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

In the Claims

In column 19, line 36, delete “herein”

Signed and Sealed this
Sixth Day of January, 2015

A handwritten signature in black ink, reading "Michelle K. Lee". The signature is fluid and cursive, with the first letters of each name being capitalized and prominent.

Michelle K. Lee
Deputy Director of the United States Patent and Trademark Office